

Nikhil D'Souza

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Education

Columbia University – New York City, NY	Dec 2023
• M.S. in Data Science • Coursework: Probability & Statistics, Applied Machine Learning, Math of Finance, Finance & Structuring for Data Science	
Purdue University – West Lafayette, IN	May 2022
• B.S. in Computer Science, Data Science, Applied Statistics (GPA: 3.84) • Coursework: Advanced Neural Projects, Data Mining & ML, Analysis of Algorithms, Neural Image Processing, Intro to AI, Data Structures & Algorithms, Large-Scale Data Analytics, Regression Analysis, Probability, Linear Algebra, Multivariable Calculus • Head Teaching Assistant of The Data Mine (1000+ students and 50+ corporate partners)	
Coursera – deeplearning.ai by Andrew Ng	2020
• Neural Networks and Deep Learning • Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization	

Experience

Vitalize – Co-founder & CTO	Feb - Sep 2022
• Built the first digital wellness platform for the healthcare workforce (see at vitalizecare.co). • Developed a clinician-centered mobile app and web-based admin dashboard using MongoDB, Express.js, React, and Node.js. • Led development team to build anonymous video coaching, dynamic surveys, and analytics dashboard.	
Fast MRI Reconstruction using Deep Learning – Undergraduate Researcher	Aug 2021 - May 2022
• Investigated the use of AI to make MRI scans up to 10x faster by reconstructing accurate images from undersampled data. • Trained a generative adversarial network with images that are Fourier transforms of 20% subsampled and fully sampled k-space data. • Utilized a GAN with a U-Net-based generator architecture and residual-in-residual dense blocks (RRDBs).	
Eli Lilly – AI/ML Intern	May 2021 - Aug 2021
• Analyzed clinical trial data to predict trial site performance and identify new indications of in-market Lilly drugs. • Used PCA, logistic regression, and XGBoost to determine if Lilly-sponsored trials should continue being funded. • Identified reasons why investigators reforecast trial enrollment estimates using TF-IDF.	
Vincere Health – Software Engineer & Data Scientist	Oct 2019 - Aug 2020
• Built a platform that uses financial incentives, behavioral nudges, and evidence-based interventions to help quit smoking and be healthy. • Developed patient facial recognition, chat messaging using sockets, and incentive engine using Redis. • Spearheaded data science capability providing actionable insights, user performance, and patient rewards.	
Saama – AI Intern	Jun 2018 - Aug 2018
• Used supervised ML techniques to identify packaging anomalies on a conveyor belt for a leading biopharmaceutical company. • Developed CNN with 99% validation accuracy and leveraged image augmentation to expand the dataset.	

Projects and Awards

Mask Autoencoder – Smart Software Hackathon (2nd Place Overall Winner)	2021
Feverbase – COVID clinical trial data platform	2020
Harvard President's Innovation Challenge – Grand Prize Winner	2020
L3Harris Technologies Scholarship – Purdue CS Awards	2019
Calculating Fatigue of Rugby Athletes – ASA DataFest (1st Place Overall Winner)	2019
Predicting Emerging eSport Celebrities – Krannert eSport Hackathon (3rd Place Overall Winner)	2018
GluClose – PennApps XVIII (2nd Place Overall Winner)	2018
Apple Worldwide Developer Conference Scholarship Winner	2015

Skills

Languages/Tools: Python, R, SQL, Tensorflow, Pytorch, Flask, React, Typescript, JS ES6, Node.js, Swift, Java, C, Unix, Git, Docker, Redis
AWS: EC2, S3, Lambda, API Gateway, Sagemaker, Elasticsearch, CloudWatch, Rekognition, Comprehend, SNS, Cognito